

ActInf Textbook Group ~ Cohort 5 ~ Session 15 (Chapter 6, Part 2) ~ 2/12/2

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 15 (Chapter 6, Part 2) ~ 2/12/2024 | 58:09

all

right welcome back everyone

it's February 12th and second discussion

of chapter 6 for cohort 5

so any thoughts or feelings on where to

begin with chapter

like any thought or quote or

[Music]

question I have

one yes

please

um I'm not so sure it's in this chapter

but can you go into more detail in terms

of uh the difference between the high

road and the low

road yes would anyone like to give a

thought on high road and low road in the

chapter

six so really getting to the recipe and

the cooking what are the high road and

the low

road

well one thought is chapter 6 is a

recipe for actually walking on the low

road because this is

where the generative model is

constructed and

specified so it kind of starts from

having

nothing I don't know how deep we um uh

have or or need to go into the food

maybe it's somewhere here but I mean

what are the ingredients what's the

kitchen what needs to be done in what

order that's the low road of building

any model there could be a chapter six
for a recipe for Designing SPM analysis
or designing linear regression analysis
as thousands of other guide books for
modeling are out there so in that sense
that all modeling
and
uh construction like this is low road
because it's actually
putting uh model down that's what's
similar about chapter 6 and another
recipe for basian
modeling and then the difference with
active inference is the connection
through the high road to the free energy
principle so that's in so relative to if
I were to to translate that um and you
know and I'm the model that I'm I'm
building is more of a a self-modeling
not trying to build a
software llm model um so in terms of
yeah the low the low road is is
Imagining the pathway and the high road
would be taking the pathway would that
be a way to contrast
them you know simp IC
form what would someone else say Yes um
Ali and then anyone else who raises
their
hand okay so uh the high road uh can we
described kind of topdown approach to
active inference which starts by asking
some uh very fundamental questions about
uh about what's required for any Cod UN
code thing to exist and what are its
necessary
properties so uh by beginning from uh a
kind of as friston himself would call it

a near existential

nihilism uh it gradually begins to build

uh some framework which can account for

any sentient behavior that agent must

possess in order

uh in order for it to possess its

integrity and to persist through time so

that's basically uh the premise of high

road but uh on the other hand for low

road it begins by asking it it begins by

the premise of uh the existence of the

mind so here uh the the existence of the

agent or rather the predictive agent is

presupposed and then it begins to

formulate or describe uh What uh kind of

mechanisms uh can realistically

describe uh its uh predictive behavior

and it's uh I don't know brain activity

uh or

rather the way that the brain act the

brain

constructs uh the inner model or uh in

know words its generative models uh so

that's basically beginning from the

premise of the existence of mind and uh

to describe uh what kind of um

mechanisms uh that particular mind

should possess in order for it to uh to

do both inference and action in a

changing environment but ultimately they

uh they achieve similar goal but uh as

uh the I mean the name of these

approaches would suggest uh only the

approach to achieve that same goal uh is

different but then again uh they

basically try to uh I mean describe the

same kind of

behavior so

you the near existential nihilism would
uh would that um would that correlate
with let's just assume they have
agency okay so uh all right so yes
agency is a kind of I mean Central
presupposition for any kind of um any
kind of thing that can
meaningfully uh navigate through its
environment and to have any kind of goal
directed Behavior so in other words
agency is a kind of synonym for a gold
directed Behavior it can be um
obviously I mean argued whether agency
really equates uh gold directed Behavior
but that's basically as far as I
understand uh that's the meaning of
agency in active inference literature or
or at least in uh the majority of active
inference

literature thank you um Kristoff then
Andrew

yeah I just want to say that for me the
um kind of a enlightening perspective on
the on the low road um is to Simply
derive the

um

uh an equation that features a free energy essentially from from Bas rule so
you can take the Bas rule um you know uh
write it in such a way that um you
substitute another distribution for the
um for the posterior distribution which
is not not generally known
um with an approximate posterior and say
like okay since I cannot know the true
posterior I'm going to approximate the
true posterior of an approximate
posterior using some kind of variational

methods and essentially using just um
the prod sum probabilities and
rearranging those equations a little bit
uh you effectively get to the point
where um you have a quantity that is
mathematically isomorphic to you know a
free energy in thermodynamics so that
that's a kind of a very mechanical
mathematical way of um teasing the the
concept of free energy out of
essentially out of nowhere right because
you start with an equation that's that's
true B base um B Rule and then you make
it slight of hand where you just simply
substitute you know Q for p and say like
this is a slightly different kind of
thing because I'm going to have a bunch
of parameters that Define what that Q
thing is and those parameters I can
somehow tweak to find how good my uh
approximate um approximate posterior is
um but because for simple um mechanical
um substitutions like that you can
derive effectively an equation that
looks like um uh free energy that's how
you just get all the way from Bas fum to
to the uh to the free energy principle
effectively so that's that's kind of
like why it's bottom up you sort of
construct it um this
way amazing thank you Andrew
oh on me

Andrew just a cute analogy I would say
that Plato would always try to start
with the high road while um Aristotle
would uh see what he's got available and
start with the low road Road and since
we're trying to implement something we

got to use the tools we have accessible
and so um chapter six is uh seeing what
we've got there and taking the low road
to uh Implement something that shows
active

awesome yeah great great comments here's

figure one two it's a nice evocative

figure also though the it's not like

there's a strict sequence or like

there's any it's like um some

subway stations but it's not very strict

and it's not like all or necessary

sufficient or anything like that um but

it is situating active inference

generative models as on one hand

implementing Bas theorem giving what

Kristoff um provided this kind of

unrecognized

relationship between statistical

principles of least action which we now

call the basy mechanics and the free

energy principle and how that's like an

extension of other physics and

information and thermal

physics and yet all all of that kind of

like Theory aside chapter 6 doesn't have

an equation

chapter 6 is like actually the sequence

although they um make clear like it's

more about a checklist um than a than an

ordered sequence here um but this is the

theory um light sequence of questions

that's actually procedural like a

catechism to work towards doing a

generative model to working in active

inference and realizing the synthetic

active inference agents

whether it's about

oneself um that's human system of
Interest or whether it's about some
other
system
um that's chapter
six it's with only looks like that one
figure and not any um of these
connections that have been brought up
which are super great and well said
about the basing
mechanics
so chapter six on the low road but if
you um build it this
way then by making an active inference
model from the low road it would
be uh you could run it in pmtip so it
would be kind
of concordant with free energy
principle and we have we have many
examples we can look at we can look at
the tutorials of
pmdp we can also talk more about the
recipe of chapter 6 but that's kind of
the big idea of
six um anyone can raise their hand with
like a or
or yes I have a question yes go for it
um yes I wrote it in the chat what is
meant by the first step the
identification of boundaries and then it
says I.E Markoff blanket
but I just want to say thank you so much
for this very um elucidating explanation
for high load and Rollo I've read it
several times and this was the best or
finally it it clicked so thank
you thank you so great
question Alie want to go for it or how

does it come into play in in the process of building a generative model yes please yeah just wrote it in the chat as well so markup blanket is a kind of a statistical boundary in the state space of the agent so in other words it can uh manifest itself as a spatial temporal boundary or not so in some cases for example in U I don't know cell boundaries or cell membranes it actually manifests uh as uh real spatial temporal boundary but it's not necessarily so uh and uh it can u i mean markup blanket can be expressed variously in different forms for different situations uh but basically it kind of provide uh a necessary statistical boundary in the state space of the agent uh in order for uh for the agent to be uh statistically separated from its uh so-called environment uh so if there wasn't any Mark of blanket delineating the this boundary then uh the behavior the internal behavior of the agent uh would be identical uh to the uh behavior of uh the environment in which it's embedded so uh yeah that's a necessary presupposition for doing any kind of active inference modeling uh around that so may I ask a follow-up question yeah sure so in the case of swarms or plasmoids where an act an individual agent is impacted by the whole group um so would the same principles apply in characterizing the agent as the group yeah that's very great question so

you see Mark blanket uh can be um can be described in uh multiple scales so for instance um even for um for so let's uh take the brain as an example uh we can draw a markup blanket around single neurons or around I don't know uh particular circuits neural circuits or around the whole brain and even even further than that around the whole organism and there are some Works uh that that have been used for um they've used active inference for modeling the whole um some of the social and cultural Behavior Uh of whole uh communities or societies so in those cases markup blankets can be drawn around a whole Community or Society so basically it depends on the system of interest uh it's not something predefined thank you I'll give one other example just because this swarm modeling topic um is a great one and to give a a specific example so when we were simulating an ant colony and there was a chapter 6 but this is the result of the chapter 6 this is the specification of the environment in a maze where there could be movement and then this is the generative model that was specified like the generative model of the nestmate was the ABCDE and the the kind of um active inference Loop that it implemented at each time step so in that sense it was an agent based simulation much like net logo or using any other agent based simulation except that the the the basian graph of the structure that it

was implementing was like the figure 4.3 graph so specification of the environment specification of the um generative model including the boundary in space and time like having the ability to receive um uh in a one square perimeter so we didn't a priori set a blanket on the colony but you could imagine if you said okay well I'm going to look down from the top and enclose another blanket around the total space that the colony encloses which might might be meaningful or might uh not be but from the modeling perspective all that was needed to specify was the nestmate but then summary statistics could be made about areas covered so there's a lot of other um analogous examples with multi-agent simulations um to a first approximation what you specify is the agents just like any other agent based model um and then um people are looking into as Ali mentioned like integrating true multiscale systems modeling like multi-agent models of neurons constituting circuits and so on Andrew and then Pros yeah just a quick note that the Markoff blanket is best thought of as a computationally uh meaningful parameter not necessarily something that you will be able to pin down in physical space so for example like if you think of your Markoff blanket it's uh associated with the sensing and action um modalities you have available um

you're even though there's physical space is going through you in a certain way and you are regenerating in terms of um your particles at a certain rate um it it's related to how you as a computational agent are associating with your environment and so uh you can think of that both from the intrinsic perspective I just described as well as if you are modeling something you get to pick what is the data going into and being processed by your statistical model and what is the data going out and being processed uh and transformed into physical uh phenomena got it thank

you thank you pros and then

Kristoff yes uh thank you uh Daniel I have a question for you like in the ant model which you showed I was imagining that if we have say multiple ant and they want to form a group by themselves uh so will you be able to create like kind of a markof model like a blanket dynamically am I correct or incorrect well few things to say on that so first the as the blankets are features of the maps not the territories like it it could be possible you could say again I'm taking a top down view of the movement and I'm going to draw little enclosures and I'm going to have a thresholding parameter for when I declare like these three to be a separate cluster so so but at that point that's not really like what the nestate is doing which this whole time hasn't

changed from just as specified so
there's a lot of higher order clustering
and and partitioning methods that aren't
exactly the base level specification of
the generative model which is often
pretty small and and simple um and then
another thing to say on it is like uh as
with many agent based model like here
all they could do was um move and uh put
down one pheromone there wasn't like an
option for them to have a second pheromone
or to do nursing or uh other kinds of
behavior
so um there's always kind of a limited
zone of of the like emergence or
phenomena that are displayed from a
simulation if it's really pruned down to
be
simple and then to make um
richer like more more whole iguana
simulations that's what is exciting
about active
inference uh one one followup question
on this uh so basically you are
designing it through policies which you
have or it's like more higher order
stuff this one the only policies they
could take there's no planning was just
moving in a grid like a grid world but
as with any active inference generative
model like chapter 6 describes all the
aspects of the model are specified the
input sensory any um higher order
beliefs that it entertains like the the
cognitive States than the affordances it
has how it does policy selection whether
it do planning whether it has any of
these other features like attention so

figure 4.3 which is basically what is
here um is just the minimal kernel Motif
in discrete time

and then there's all of these kind of
like features and this is a very simple
framework with the goal as most of the
published models

are and then um that's why there's so
many opportunities now to connect the
different systems of Interest with the
colonel with the motifs that are
arising um okay Francis yeah thank you
Francis oh sorry Kristoff and then
Francis then

Tanner

uh I just want to add one thing to the
um Mark of blanket concept
um slightly different perspective that
uh comes from something that um Steven
walr is working on the Observer Theory
which is he's kind of rediscovering very
similar thing to active inference but on
slightly different um um basis so he
talks about what's necessary for for
something to be on Observer and in some
sense you need to have more computation
going um going on on the inside of the
Observer and you know that's that's
going on on the outside what what is
being observed so this Mark of blanket
is the statistical definition of a mark
of blanket to me is is kind of um very
confusing right like there's this uh
statistical what does the statistical
Independence really mean well it really
means that you you know the the The
Observer or the agent is going to do
something internally that you can't

readily predict from just looking at it
without um it actually responding
somehow to the event so it has to you
know um some information is coming in it
has to process it in some way and it you
know it will react react in the
eventually in the environment and as
long as it's doing enough computation on
the inside then um
the uh it will basically become
statistically independent because you
can no longer using simple statistics or
any kind of Statistics really model uh
what is it really um what is it really
doing uh
internally and these things kind of seem
to arise naturally as uh any kind of
complex systems uh interact so so these
are kind of boundaries that on one hand
we can we can draw them somewhat
arbitrarily sometimes on the other hand
they're they're kind of naturally
arising in interactions of any kind of
um any kind of computational
systems I hope it's this is very helpful
for me but um I'm not sure if it makes
any sense in this context here yeah it's
awesome like you're you're right they
can absolutely be proposed um out of
nothing like you could just draw
arbitrary base graph and just say here's
the blanket but they also do arise
naturally seemingly um or maps with this
character describe complex systems um a
simple example of which being space um
or time like things kind of propagating
outwards in space and things not
teleporting through space like a bubble

expanding outwards not like skipping a shell because it's like propagating locally in space so that kind of captures that property too as well as the I guess as wol from is looking into more of the computational parts of like measuring and and observing um Francis then Tanner and Ali hi

so uh my first stab at understanding marov blankets was uh from my programming background I thought okay so this is like a bit like having a function or an object where you pass parameters in which might be the goals and then leave it to get on with this and then maybe it'll some if something goes wrong it'll send an exception up to uh so that can go up the perception the exceptions go up the perception tra uh

hierarchy

um and

then I just been I picked up the uh references to the environment and I'm starting to think of as a mark of blanket as being something which uh because things are hidden from the environment they're not part of the environment therefore they can have a is the internal State and it can be the internal state which as an to the intent that to the extent that it's an agent the internal state that it is trying to preserve

um so it's kind of what's at stake what it

is
um at it
being uh I guess anything which has a
barkoff
blanket um
actually what what is the answer to that
question what is it that can have a
markof
blankets uh an active
entrance okay thank
you thank you

Tanner okay so yeah I have a I do have a
question that I want to ask but first I
um I want to respond to the last
participants query um I think the the
issue that a lot of people have when
trying to understand what marov blankets
and marov boundaries are is they like
make it too agential in the sense of
like the words that they use just become
agent identified
like the way that I understand it is
that marov boundaries are agents right
so so we say but it's important to
understand that marov blankets aren't
actually extent in like material reality
they're just ideas right they're like
this is like this is the boundary um so
you know it it you can sort of
understand this if you like think about
it and you're like you know I so I see a
spider right and I'm like that's a
spider that's that's an
individual but is it like you just kind
of made that definition right you you
assigned the you drew the Line in the
Sand right you you know circled The
Spider and we're like that's that's the

entity right there that's the agent with
identified active awareness doing the
inferential you know

stuff it's kind of arbitrary so marov
blankets are like

defined in this uh you know more hor
horis way like they are they are
emergent properties that do exist like
they they do they exist in a qualitative
sense um you know most of the time I'm
existing as as a person in a body but in
a state of meditation you can sort of
start to exist with a larger or just
different marov blanket that includes
maybe other people the room Etc so
that's that's all I wanted to say about
that it's not

it's not one to one right there's

there's there's this there's a whole

Spectrum to these things um anyway my

question was about spiders so you uh

were you have these models um that you

were presenting about ants and my

curiosity is uh ants have this as I

understand largely two-dimensional kind

of experience where it's like left right

you know they're not really moving up

down it's it's very much so just like

follow the fairmon trail um and you can

correct me if I'm wrong because you know

much much more about this than I do but

my my question was like if you were

trying to extend that uh this model that

you've created to a spider which is

something that exists in in a very

three-dimensional way and even in in

some sense in like a four dimensional

way because the whole concept of

building a web is like working with a plan that is a long-term plan to catch and exist right it builds a home for itself that's not something a drone ant is doing I don't think so just curious what you kind of think about that idea if just take like a a classical like field spider as an example um and and I would I would love to to hear your thoughts on that in comparison to your model

so okay just very briefly um the Markov blanket is definitely not associated with just the agent environment interface like this node is a blanket between X and Y so the markup blanket can be identified between any nodes but one kind of interest is this agentic and and definitely it it's like very um Observer dependent like all observation is how agentic different boundaries are seen as and um to the spider I mean this is what chapter 6 is about please look through chapter 6 and mega spider model and then next week let's look at it and we'll further along and maybe we can even you know go from there but that's exactly what chapter 6 is about yeah um do you have a recommendation do you have a recommendation of a particular species I should look at or I can find one I I mean from the beginning of when you're feeling curious about the phenomena on through the active inference model you make just document it and then it'll be done okay cool thank you Daniel I appreciate it thank

you uh so uh little bit I think u a more sophisticated uh version of Mark of blanket I'm not sure if it's directly uh inspired by the concept of markco blanket or not may or may not be but uh there's this uh important paper by Michael 11 one of its um I believe one of the most highly cited ones the computational boundary of a cell from 2019 uh in which he develops uh the concept of scale-free cognition and the idea is that there's this computational boundary or as um as he describes it in this paper a kind of light cone a cognitive light cone that can Define uh the scale of the agent's cognition um I mean through various scales and various limits of the cognition so it's not restricted uh to uh even a single scale of interest uh it can be uh modeled u in a in a true hierarchical way to include uh various um scales and various limitations of cognition even if we want to kind of investigate the interaction uh between uh or among um the various scales of cognition and the other uh related um conception uh was developed by uh David kraar at all and their 2020 paper the information theory of individuality in which they again uh describe a kind of U information theoretical conception of uh what it means to be a biological individual so uh this information theory of individuality uh enables uh the identification of uh of the agents at uh any level of organization I mean from uh even molecular uh

molecular level to even the cultural ones uh so uh these recent concepts uh can be seen as kind of update on the or we can look at them as kind of Mark of blanket 2.0 but I believe uh familiarity with these newer Concepts can uh even help us to understand uh the nature of Mark of blanket uh I mean uh more clearly uh than just looking at the Active inference literature because I agree that this concept can seem a little bit vague uh for newcomers uh or even for experts as well uh so I highly recommend those two papers as well thank you Lee yeah a few comments on that like the Mark of blanket concept comes from the causal Bayesian modeling literature and then there's various extensions and kind of analogous the the blanket index all these other things um to kind of uh I mean bring it to a more low road building does anyone want to look at the questions or share some of their modeling experiences or we could look at the PDP tutorials because the end of chapter 6 not not like it only has to be about this chapter but the end of chapter 6 um doesn't necessarily leave one um further along or scratching their head about these uh like multiscale systems uh except through the construction and often like going through and talking through and building through a system gets really far and then even the theory questions from

there are more
informed
so it's not like the theory has to be
resolved
like
before one builds anything okay got it
that's useful like the PDP tutorials are
are we or or does anyone want to look at
or what would be helpful in our last
bit I I think like yeah we should
if everybody agrees
pmdp yeah um so
like let's do it there are some uh I
think a model stream 7.1 and two with
Connor Hines and Daphne dcus um like but
those are long videos and um also
contain a lot more of like
the the background on
pmdp um so I just put it into the
six okay so um I
mean the big picture on what's going to
happen here and how this is different
than chapter six this is not um this
notebook itself is not um a supplemental
dish to chapter six however with some
very slight tweaks it could be so just
to kind of set set the scene like
chapter six is a sequence of questions
it's a catechism basically for going
from your curiosity
to the constructed model so what system
are we modeling are we modeling one
spider three spiders three spiders in a
web three spiders in two webs five
spiders on N webs two trees you know
just that is not a trivial question what
are you modeling and so figuring this
out is what they have as the first

question then

what's the appropriate form of the

generative

model one big distinction being like

figure 4.3 discrete time or continuous

time and all the strengths and

weaknesses of these different um

techniques um also pmdp is like only for

discrete time models right now but

there's other approaches for continuous

time modeling um then what's the

structure of the generative model is it

an estimate that can move two two

squares forward or one Square the side

or this way diagonally or how many

phermones and how many uh does it have a

tension and if it doesn't have those

features it's not going to have them if

it doesn't have three-step time Horizon

planning your generative model will not

have that feature like if it doesn't

have planning in the code it will not do

planning now that doesn't mean it won't

give behavior that looks very

interesting in other ways but actually

that's a feature because you can say

through the fully synthetic nature of

the model it doesn't do planning and yet

this is happening um or it doesn't have

a preference for this and yet this is

happening and then what they leave for

last is like how to set up the

generative process however um as Ali and

uh asked Maxwell ramstead on in the

guest

stream they have kind of moved away that

group and of modelers not saying this is

the the only the best way they've moved

away from distinguishing so much between generative model and process because in the end they're co-defined and in like complex systems it's not so simple it's like which one's the agent which one's the environment it's like well there's two agents communicating so you know why not just call it all the generative model um okay so chapter six gets us to writing the pseudo code basically that we're going to look at the python code for

um what they were going for here their goal was um you know if if we were to apply the chapter 6 question is to model an agent that moves in this um 3x3 maze grid world and has a preference for a certain location and you can copy the the um code this package that they have written is called PDP um and it also is bringing in just a few math packages this box this is tutorial one this box has some kind of um just purely visualization based functions like plotting likelihood this is not like an active inference

box um also as some of my friends have remarked from computer science is like there's some pros and cons to notebooks on one hand they can be be copied really easily and run in the cloud like on a Chromebook but at the same time it can like lead to a lot of like okay I thought we were going to do active inference but then no it's like this other thing is happening first so it's

not always clear what's what's happening
even what because it's so
linearized
okay here they're just showing what this
function does and plotting a categorical
distribution
so it's it's kind of built it's like a
little step by step this is not building
the agent yet then from just doing this
is like a three-sided coin
flip now the other kind of key piece of
the Bas formula is going to come into
play which is conditional
probabilities and they show doing some
conditional probabilities just to show
in PDP like what we're doing is
probabilities and conditional
probabilities like the The Joint
distribution of the nesme is just the
joint distribution of a b c and d that
totally describes the nestmate there's
no other information about the nestmate
and then conditional probabilities
become really important because like
we're often interested in conditional
distributions like the distribution of
action conditioned upon dis
observation other things like that okay
so that that was all like kind of
preliminary
PDP utilities and just demonstrating the
syntax kind of now they're going to go
into grid
World
okay the grid is specified it's a 3X3
grid that's specified as this um set of
ordered
pairs and then it's visualized with

their graphing

function

so that's the generative process that's

basically the environment is being

specified now just like for other

discrete time active inference agents

what there is to specify is a b c and

d so just to the figure uh or

chapter 7 where it goes which is the next

chapter which goes into more detail on

the discrete time model so a b c and d a

b c going into μ G here's C because

policy distribution is conditioned upon

preference and habit so it's just kind

of getting wrapped into this G

visualization a b c

d first they generate a a is the

conditional distribution that is

basically mapping between observations

and hidden States

so μ in this world each of these

locations is going to μ just it's

basically labeled like with its location

so it has like a perfect μ

GPS μ so the dimensionality of a is set

the dimensionality of a is basically

like observations mapping to Hidden

States it's a matrix that describes that

mapping and so they're the same in this

case then it's made out with zeros just

to initiate it and then it's filled on

the diagonal with ones which corresponds

to perfect mapping of States if the a

matrix were flat there'd be no

information about the world coming in

ETC here it's the identity Matrix so

it's just a noisess

mapping that's what it looks like so

here white square is one likelihood
black square is zero likelihood and then
it's mapping from the nine observations
to the nine hidden States and it looks
exactly this like if you see a three it
always maps to being in state
three here and then um they make a noisy
a with some different noise patterns so
here observation zero is ambiguous
whether it maps to 01 or three so
following in the grand SPM tradition
even very large generative models can be
specified like with these grayscale heat
Maps it's actually really
cool two ambiguous locations now six is
ambiguous as well okay so that's a so we
have a few different a matrices as with
other modeling it's never just about one
one perfect model that describes like
the spider it's about like having
multiscale portfolios of a matrices and
composing them and and displaying like
like this one has this behavior and
think it's cuz this but actually here's
this other model that shows that it's
not so it's it's not just about making
one it's about like doing a lot of
models and sweeping across them now they
go on to b b is the hidden State
transition um each slice of B is going
to describe the mapping from one world
State at past time point to the next
time point and then different slices of
B get selected based upon the um policy
selection so B is the critical variable
in active inference this is where the
action dependent World model is
specified and like this is really the

variable that explodes a lot with model complexity because the D variable is just a prior on world States so it's just a vector that kicks the chain off one time with like an initial Distribution on room temperature and a is a sens sensory mapping from thermometer reading to the hidden State and then C is just a preference over thermometer readings so all of the work in the model with the action dependent causality all of it is in B it's basically the only place they can go just in practice so bees created these are the affordances they're called actions in this script um there are five of them up down left right stay um a few pieces are uh just visualized and just checking through that uh they can advance the next state and then um like take the action and move from here and then move right one square to here then move down to here so that's B then C they zero it out C has the same length as observations because it's a distribution over observations and here it's given a value of one for eight and zeros for the rest and then D has a length of the number of world States and it's a distribution of our world States and so here it starts 100% of the time state

zero and that's just how it is so that's
that's the ABCD
specification that's all of the so at
every single time Point all of the all
that the agent simulation is one agent
or multi-agent it's just a b c and d for
each agent in this most simple
example then they make some um functions
in first state that's like the making
part and this just describes the pseudo
code for going from the
observations combining that with a prior
and returning um Q of hidden state that
we discussed Q of X earlier
today they do a Time step of just the
sensemaking
inference including with the ambiguous
priors so that's the sensemaking part
then action selection this is the action
part um some more functions are
introduced pretty cool to see the uh
entropy and the K Divergence just toine
so
shortly they just kind of replot things
and just kind of refresh where it's all
at change the preference to five
add a few um print
statements
compare the expected free energy for
moving left versus moving right
classical approach you have your
affordances that are being considered
here just those two just for clarity in
the tutorial and then EF G is calculated
for those
two and then a softmax is applied such
that the units of G are normalized to
probability

distribution and then here is the whole um grid world and then they all wrap it together and then they just output how it happens and then there's a few more details and then they add planning at the end but everything above this point is no planning and then from this point you you start to see the planning I hope that's useful um and that people make also other tutorials or maybe contribute edits to PDP because there's a lot of experience with modeling that people have that that could really help and it's it's um really fun with the theory but also it's a very specific modeling possibility and we have the um there's there's so for for those with computer experience many of these will be accessible um for people with less coding experience like it will be really helpful to figure out and like to learn what approaches um are exciting for you because this is the low route a low route it's a simple model but it it kind of is it helps calibrate and Benchmark like if this is um a big pill then you know simulating 500 social agents with memory is probably a really challenging project for you if this is very simple and understandable then you can probably

start to Benchmark what kinds of models
you'll be able to understand or or at
least um iterate on starting
tomorrow okay any questions in the
last five minutes or thoughts on like
this modeling topic as that's kind of
six

yes prau and anyone else uh yes I was
actually I think uh first thing I what
I'll have to do is I'll have to just uh
do this tutorial once like it's very
nice you have given a brief summary of
it so it'll be very helpful if you do it
today or tomorrow um and I was I was
thinking like if if the if the Matrix is
what you mentioned over here if that is
like we have various con Concepts like
various class data structur we know very
well so similarly if you know this
matrices quite well then I think things
will just flow that's what I understand
out of

it right thank you yes totally and also
um the the model presented here is a
pretty um common

PDP that's what gives the continuity
with the computer science and the base
graph and everything like that that it
is just a PDP

basically and and yet it's um about that
Motif and then about understanding it in
the single agent case and then just kind
of building from there so like a lot of
the the work and the challenge in
application look forward to everyone
also sharing their experiences on this
is like even building one agent and then
there's like the just like Crea the

environment with n but also people are making amazing toolkits to help facilitate multi otive simulations so just like everyone keep you know learning and exploring and the tools really are getting better fast

Susan um so I'm um my head's around how to integrate this with the Constructor law and I'm wondering where um degrees of freedom show up in this particular chapter you pointing to

yes in the narrow not necessarily in the software but I in the real world oh I mean the chapter describes several of the mega degrees total degrees of freedom like okay what is the system we're modeling and who are we doing it for and why I mean these are the ultimate degrees of modular Freedom h and then these distributions have their own statistical degrees of freedom okay so it really is the construct of freedom is is kind of underlying it's kind of you know under the soil you might say of the uh of the blanket so the perception of someone's Freedom may actually correlate with where they see their boundaries where they see their Market boundaries I almost want to say say that um if I were to transfer translate the the market the markof blankets you know to a you know a okay metaphor it's not perfect but an okay metaphor it would almost be the

protective boundaries that we um that we
perceive in the world which has a lot to
do with you know our perception of
Freedom would that
be however you feel
all right yeah that trying to stay with
the rigor um say I don't want to use
something that doesn't it's completely
off like I love that you added it I
don't know if they talk about Freedom at
all in the book so in in the letter of
the book no but I really like what you
added so okay thank you thank you
everyone peace see you next week also
feel free to do the tutorials add
questions add notes and like your
experiences if you were familiar with
the code or not and all of this could be
really fun so see you next time bye
thank you everyone thank you that was
great awesome bye